

Friday-03-07-2025 combustion Example and CEA software
Section 6

Example: Q-13 from chapter 2 of the textbook.

A mixture of air and $\text{CH}_{1.9}$ undergoes complete combustion. Twice as much air is present as needed to burn the fuel. (200% theoretical air). How high is final temperature.

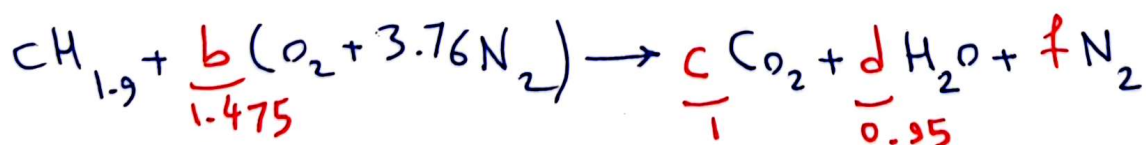
($T_R = 25^\circ\text{C}$, use the below table)

Species	$\bar{h}_f(\bar{Q}_f) \frac{\text{kJ}}{\text{kmol K}} @ 298 \text{ K}$	$\bar{c}_p \frac{\text{kJ}}{\text{kmol K}}$
$\text{CH}_{1.9}$	9358	
$\text{CO}_2(\text{g})$	-393522	51.9
$\text{O}_2(\text{g})$	0	34.0
$\text{N}_2(\text{g})$	0	31.6
$\text{H}_2\text{O}(\text{g})$	-241827	40.6

$\bar{Q}_f$ ($\bar{h}_f$) is the enthalpy of formation @ 298 K.

It means that some heat is necessary to create that species. The extra heat increases its temperature.

Solution: First, write stoichiometric reaction:



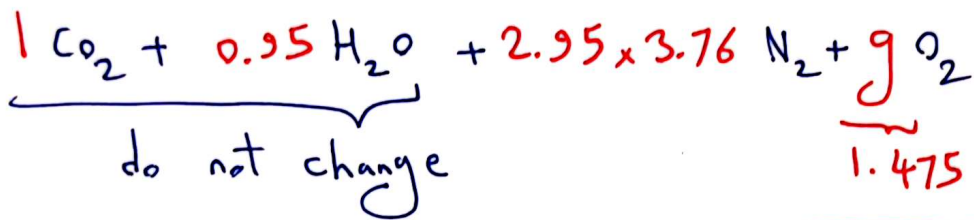
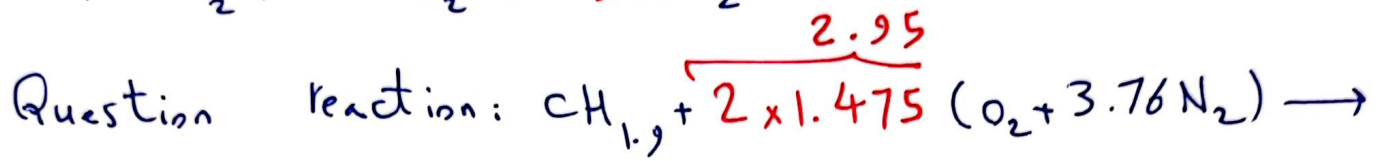
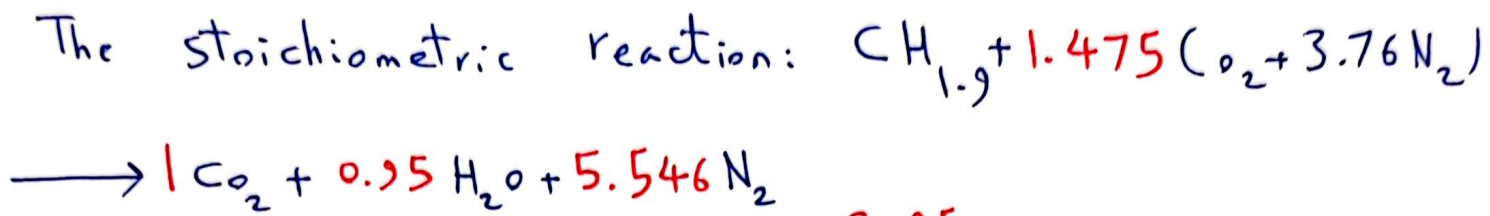
Balance order: C-H-O-N

C: $\boxed{1=c}$

H: $1.9 = 2d \rightarrow d = 0.95$

(P-1)

O: $2b = 2 \times 1 + 0.95 \rightarrow \underline{b = 1.475}$ N₂: $1.475 \times 3.76 = f = 5.54$



O: $2.95 \times 2 = 2 + 0.95 + 2g \rightarrow g = 1.475$

We balanced the real reaction, to get product temperature
use: $\bar{H}_P = \bar{H}_R$ or $n\bar{h}_P = n\bar{h}_R$

Note: Enthalpy consists of enthalpy of formation @ 298 K plus extra enthalpy for being at higher temperature.

For products: $\bar{H}_P = n_{\text{CO}_2} \left[\underbrace{\bar{h}_{f_{\text{CO}_2}}}_{\text{enthalpy of formation at 298 K}} + \underbrace{\bar{c}_{p_{\text{CO}_2}}(T_P - 298)}_{\text{additional enthalpy needed to bring CO}_2 \text{ to } T_P} \right]$

$+ n_{\text{H}_2\text{O}} \left[\bar{h}_{f_{\text{H}_2\text{O}}} + \bar{c}_{p_{\text{H}_2\text{O}}}(T_P - 298) \right] + n_{\text{N}_2} \left[\cancel{\bar{h}_{f_{\text{N}_2}}} + \bar{c}_{p_{\text{N}_2}}(T_P - 298) \right]$

$+ n_{\text{O}_2} \left[\cancel{\bar{h}_{f_{\text{O}_2}}} + \bar{c}_{p_{\text{O}_2}}(T_P - 298) \right]$

Let's write the values in enthalpy of products:

$$\bar{H}_P = 1 \times [-393522 + 51.9 (T_P - 298)] + 0.95 \times [-241827 + 40.6 (T_P - 298)] + 11.092 [0 + 31.6 (T_P - 298)] + 1.475 [0 + 34 (T_P - 298)] \Rightarrow \boxed{\bar{H}_P = -623257 + 491.13 (T_P - 298)} \quad (A)$$

Now, let's go to reactants, note, reactants are at 298 K

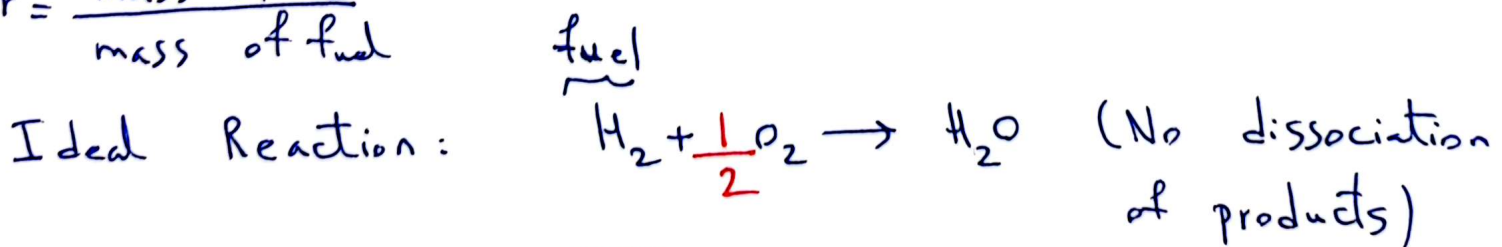
$$\bar{H}_R = n_{CH_{1.9}} \left[\bar{h}_{f, CH_{1.9}} + \cancel{\bar{c}_{P, CH_{1.9}} (T - 298)} \right] + n_{O_2} \left[\cancel{\bar{h}_{f, O_2}} + \cancel{\bar{c}_{P, O_2} (298 - 298)} \right] + n_{N_2} \left[\cancel{\bar{h}_{f, N_2}} + \cancel{\bar{c}_{P, N_2} (298 - 298)} \right] = n_{CH_{1.9}} \times \bar{h}_{f, CH_{1.9}} = \boxed{1 \times 9358} \quad (B)$$

$$\bar{H}_P = \bar{H}_R \rightarrow A = B \Rightarrow \text{Solve to get } \boxed{T_P = 1586.082 \text{ K}}$$

This is adiabatic flame temperature!

Example: Consider H_2 & O_2 reaction. Find "r" stoichiometric.

$$r = \frac{\text{mass of } O_2}{\text{mass of fuel}}$$

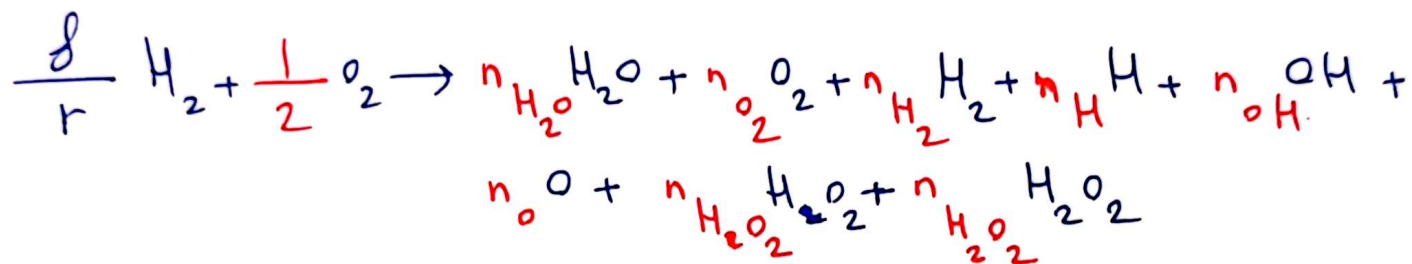


$$r)_{st} = \frac{\frac{1}{2} \times 32}{1 \times 2} = \boxed{8}$$

If $r < 1 \rightarrow$ Some H_2 remains unburnt

If $r > 1 \rightarrow$ Some O_2 remains unburnt

Now, consider red reaction between H_2 & O_2 , which gives:

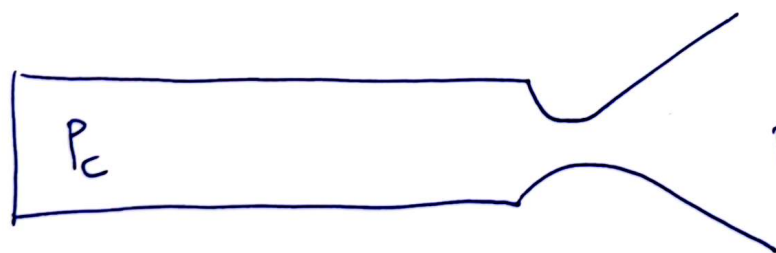


To obtain number of moles of products, use CEA software

If $r = 1 \rightarrow \frac{f}{r} = 1$, $r \equiv O/F$ in CEA software

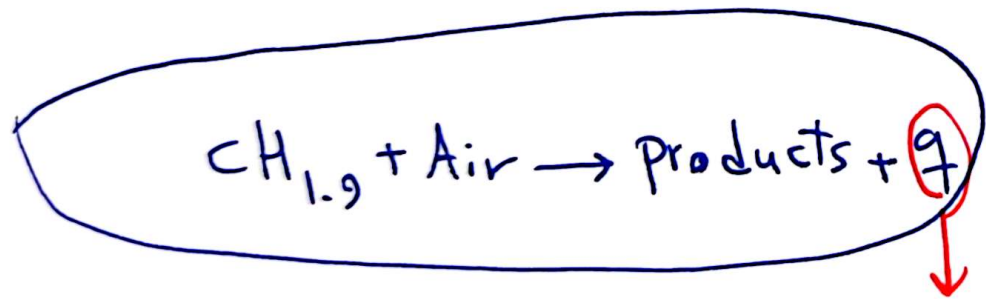
CEA (NASA combustion & gas dynamics analysis software)

It considers thrust chambers

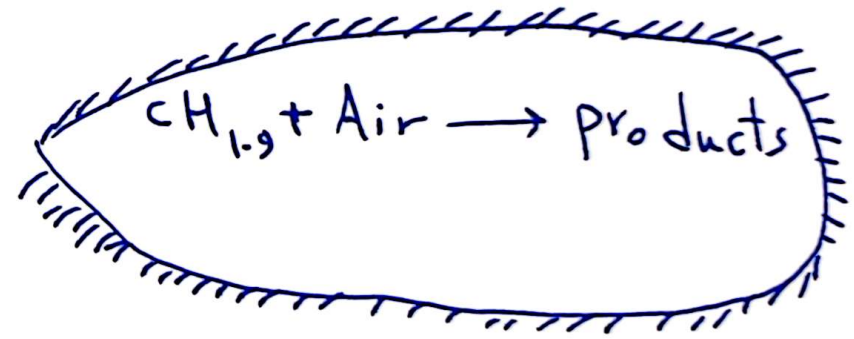


P_e $\frac{P_e}{P_c}$ or $\frac{A_c}{A^*}$ $_{sup}$
are needed by CEA

Before proceeding to compute flame temperature, let me define what is the physical meaning of adiabatic flame temperature:



Non-adiabatic
heat goes out



adiabatic

As heat can not escape,
products become hot, they
reach to "adiabatic flame
temperature"